

1 Exponential Models

1.1 Solving an Equation with an Exponent

In this section we shall explore how to solve equations of that form $3^x = 27$. This equation is nice enough to see that $x = 3$ works since $3^3 = 3 \cdot 9 = 27$. However, we also see that $27 = 3^3$. Thus, in substituting this form of 27 into the given equation, we find that $3^x = 3^3$, suggesting that $x = 3$ is the solution. We shall explore this latter more methodological approach for solving exponential equations.

We first recall some properties of exponents that will be used throughout when solving exponential equations.

Theorem 1. *Let $a > 0$ be a real number. Then, for any x and y ,*

1. *If $a \neq 1$ and $a^x = a^y$, then $x = y$.*

2. $a^x \cdot a^y = a^{x+y}$

3. $\frac{a^x}{a^y} = a^{x-y}$

4. $(a^x)^y = a^{xy}$

5. $a^{m/n} = \sqrt[n]{a^m} = (\sqrt[n]{a})^m$

We shall show by example how to solve exponential equations. However, the general scheme is as follows

1. Clear any fractions by using negative exponents.
2. If possible, rewrite everything on both sides of the equation in the same base.
3. Combine exponents in possible.
4. Compare the exponents and solve as usual.

If you have seen logarithms already, one can use logarithms in everything that follows, especially if one cannot rewrite both sides of the equation in the same base.

Example 1. *Solve for x in*

$$\frac{1}{5^{10x}} = 5^{20x+5}.$$

The first thing we need to do is clear the fraction $\frac{1}{5^{10x}}$, by which we mean, we use

$$\frac{1}{5^{10x}} = (5^{10x})^{-1} = 5^{-10x}.$$

Thus, we hope to solve

$$5^{-10x} = 5^{20x+5}.$$

But, we may now just compare the exponents and solve

$$-10x = 20x + 5.$$

Thus, $x = -\frac{1}{6}$ is the solution. $\triangle$

Example 2. *Solve for x in*

$$36^{7x} = 6^{4x-2}.$$

The idea here is that we need to first rewrite everything in the same base if we can. That is, we need to somehow change the 36 into a power of 6. But, we know that $36 = 6^2$, and so, we have

$$36^{7x} = (6^2)^{7x} = 6^{2 \cdot 7x} = 6^{14x}.$$

Thus, we need to solve

$$6^{14x} = 6^{4x-2}.$$

But, this is a matter of solving

$$14x = 4x - 2,$$

and so, $x = -\frac{1}{5}$ is the solution. △

Example 3. Solve for x in

$$10^x 10^3 = 1.$$

There are two ways to do this problem. Either we can divide through by one of the factors or we can combine exponents. We shall combine exponents here. We find that

$$10^x 10^3 = 10^{x+3}.$$

Next, we need to rewrite everything in the same base. We recall that $a^0 = 1$ for all $a \neq 0$, and so

$$1 = 10^0.$$

Therefore, we wish to solve

$$10^{x+3} = 10^0;$$

i.e., we wish to solve $x + 3 = 0$. Therefore, $x = -3$ is the solution. △

Compound Interest

In this section we concern ourselves with the mathematical model of compound interest; however, since this is a math course and not an economics course, we shall focus primarily on the practice of the math involved rather than the theory. Thus, we shall mostly deal with defining terminology without theory, and then seeing examples done.

Definition 1 (Compound Interest). Suppose a principal P earns interest at the annual rate of r , and interest is compounded m times a year. Then, the amount F after t years is

$$F = P(1 + i)^n = P \left(1 + \frac{r}{m} \right)^{mt}$$

where $n = mt$ is the number of time periods and $i = \frac{r}{m}$ is the interest per period.

Recall that the principal is what is deposited, the interest is how much is accrued after a certain period of time, and we say the interest is compounded if several time periods have passed for which interest is accrued on itself.

For convenience, we have the following cases that we may encounter listed here.

1. If interest is compounded yearly, $m = 1$.
2. If interest is compounded quarterly, $m = 4$.
3. If interest is compounded monthly, $m = 12$.
4. And so on.

Example 4. Suppose \$1000 is deposited into an account that yields 9% annually. Find the amount in the account at the end of the fifth year if the compounding is

- (a) annually,
- (b) quarterly,
- (c) monthly.

To solve this, we just need to apply the above definition with the correct values of m . Note that since we wish to find the amount at the end of the fifth year, 5 years will have passed and so $t = 5$. Next, since \$1000 is being invested, we have the principal is $P = 1000$. Lastly, since the account yields 9% annually, we have that $r = 0.09$. We note that we are to find F in the definition of compound interest.

- (a) Since the compounding is done annually, we must take $m = 1$. Thus, we apply the definition and find

$$F = P \left(1 + \frac{r}{m}\right)^{mt} = 1000 \left(1 + \frac{0.09}{1}\right)^{1 \cdot 5} \sim 1538.62$$

is the amount in the account after 5 years have passed.

- (b) Since the compounding is done quarterly, we must take $m = 4$. Thus, we apply the definition and find

$$F = P \left(1 + \frac{r}{m}\right)^{mt} = 1000 \left(1 + \frac{0.09}{4}\right)^{4 \cdot 5} \sim 1560.51$$

is the amount in the account after 5 years have passed.

- (c) Since the compounding is done monthly, we must take $m = 12$. Thus, we apply the definition and find

$$F = P \left(1 + \frac{r}{m}\right)^{mt} = 1000 \left(1 + \frac{0.09}{12}\right)^{12 \cdot 5} \sim 1564.68$$

is the amount in the account after 5 years have passed.

△

This example shows how much one will accrue in interest given a specified amount deposited. What if we wanted to know how much one should invest to return a specified amount? This is essentially how we define present value.

Definition 2. Suppose an account earns an annual rate of r and compounds m times a year. Then, the amount P , is called the **present value**, needed currently in this account so that a future amount of F will be attained in t years is given by

$$P = \frac{F}{\left(1 + \frac{r}{m}\right)^{mt}}.$$

Example 5. Suppose you want %20K after 18 years, and you've found an account that earns 9% compounded quarterly. How much money should you set aside?

Since we are compounding quarterly, we have $m = 4$. Since the account earns 9%, we have $r = 0.09$. Since we wish to determine how much money to invest in order to earn 20K in 18 years, we have $t = 18$. Lastly, since we want to earn 20K, we have $F = 20,000$. Thus, applying the definition of present value, we have

$$P = \frac{F}{\left(1 + \frac{r}{m}\right)^{mt}} = \frac{20,000}{\left(1 + \frac{0.09}{4}\right)^{4 \cdot 18}} \sim 4029.69.$$

Thus, under the given conditions, to earn 20K after 18 years, we need to invest approximately 4029 initially. △

1.2 Continuous Compounding

Here we see what happens when we so called compound continuously. The question we are essentially asking is, What if we compound not discretely (e.g., monthly, quarterly, daily), but rather continuously (think infinitesimally)? It's a wonderful result that a constant (like π) pops up in this theory. The constant is e and is approximately equal to 3.

Definition 3 (Continuous compounding). If a principal P earns interest at the annual rate of r , and interest is compounded continuously, then the amount F after t years is $F = P e^{rt}$.

Example 6. Suppose \$1000 is invested at an annual rate of 9% compounded continuously. How much is in the account after

(a) 1 year.

(b) 3 years.

(a) Like the previous problems, this is just a matter of applying the above definition. We have $P = 1000$, $r = 0.09$, and $t = 1$. Thus,

$$F = Pe^{rt} = 1000e^{0.09 \cdot 1} \sim 1094.17$$

(b) Here $P = 1000$, $r = 0$, and $t = 3$. Thus,

$$F = Pe^{rt} = 1000e^{0.09 \cdot 3} \sim 1309.96.$$

△

The last thing we remark on is the present value for continuous compounding. Note that the process is almost identical to the discrete compounding above.

Definition 4 (Present value for continuous compounding). Suppose an account earns an annual rate of r , and compounds continuously. Then the amount P , called the present value, needed presently in this account so that a future amount of F will be attained in t years is given by

$$P = Fe^{-rt}.$$

Example 7. Suppose you want \$20K after 18 years, and you've found an account that earns 9% compounded continuously. How much money should you set aside?

Since the account earns 9%, we have $r = 0.09$. Since we wish to determine how much money to invest in order to earn 20K in 18 years, we have $t = 18$. Lastly, since we want to earn 20K, we have $F = 20,000$. Thus, applying the definition of present value, we have

$$P = 20000e^{-0.09 \cdot 18} \sim 3957.97$$

Thus, under the given conditions, to earn 20K after 18 years, we need to invest approximately 3957 initially. △

2 Logarithms

In this section we explore a function that essential “undoes” what exponentiation does.

Definition 5 (Logarithm). Let $a > 0$, $a \neq 1$. If $x > 0$, then the logarithm base a of x , denoted $\log_a x$, is defined as the number y such that $x = a^y$; i.e.,

$$y = \log_a x \text{ if and only if } x = a^y.$$

Before we see examples, let us remark on two properties of the logarithm first.

Theorem 2. If $a > 0$, $a \neq 1$, then $a^{\log_a x} = x$ for all $x > 0$. Moreover, $\log_a a^x = x$ for all x .

Example 8. Compute the following

1. $10^{\log_{10} 4}$
2. $24^{\log_{24} 15}$
3. $\log_{\pi} \pi^5$
4. $\log_3 3^{-25}$

△